

Evaluating the Cutoff Frequency of an RC low-Pass Filter

Students of the academic year 2024/2025

09/06/2025



UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

**MEASUREMENTS FOR TELECOMMUNICATION
SYSTEMS by Prof. Francesco Lamonaca
A.A. 2024/2025**

This work is part of the project:

EXTENDABLE

NETWORK OF **EXTENDED REALITY-ENABLED**
LABORATORIES FOR REMOTE PRACTICAL TRAINING

Web-site: <https://extendable.dieti.unina.it/>



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Ministero
dell'Università
e della Ricerca



Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

**The Research Team involving Units from:
University of Calabria, Naples Federico II, and Sannio
is proposing the creation of a network of laboratories**



Process

Goals



Theory behind



How to create the filter?

Frequency Response
of a 1st-order LPF

Uncertainty Model



Instruments



Questions?

What are the instruments?

How to link them?



Procedure



Sinusoidal Signal

Calculate the gain ratio

**Change the frequency
[0 Hz – 300 kHz] in steps
of 10 kHz**

**Experimental
Evaluation**

Uncertainty

Cut off frequency



Conclusion



**Experimental Vs
Theoretical results**

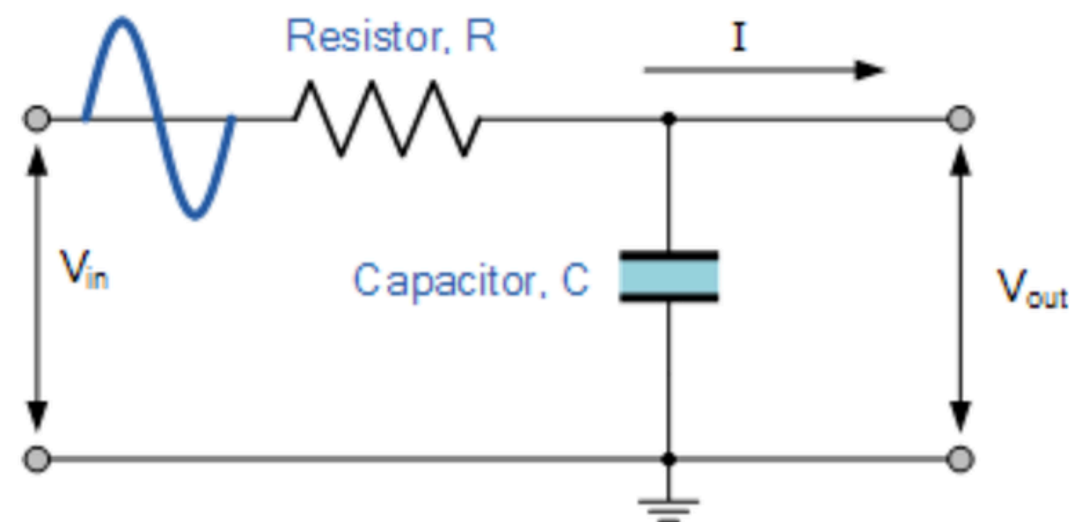
Hard and soft skills

Goals

**Evaluating the Cutoff Frequency of an
RC low-Pass Filter**

Theory behind

How to create the filter?



Circuit

- Connecting together in series a Resistor with a Capacitor
- Known generally as a “first-order filter”

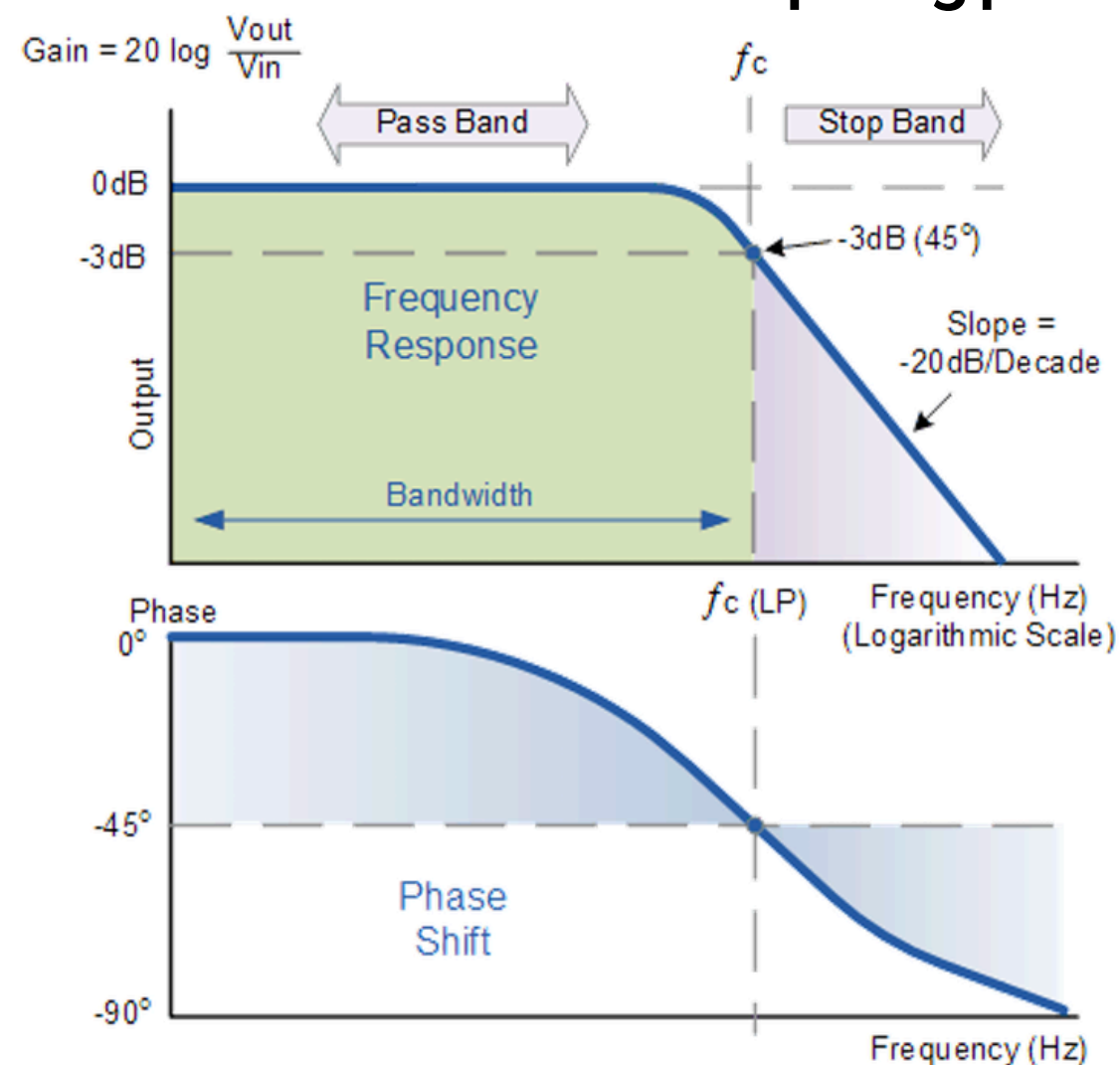
Why first-order or single-pole?

- Because it has only “one” reactive component, the capacitor.

Theory behind

Frequency Response of a 1st-order Low Pass Filter

Cut-off Frequency point



The Bode Plot shows:

Nearly flat for low frequencies and all of the input signal is passed directly to the output, resulting in a gain of nearly 1, called unity, until it reaches its f_c

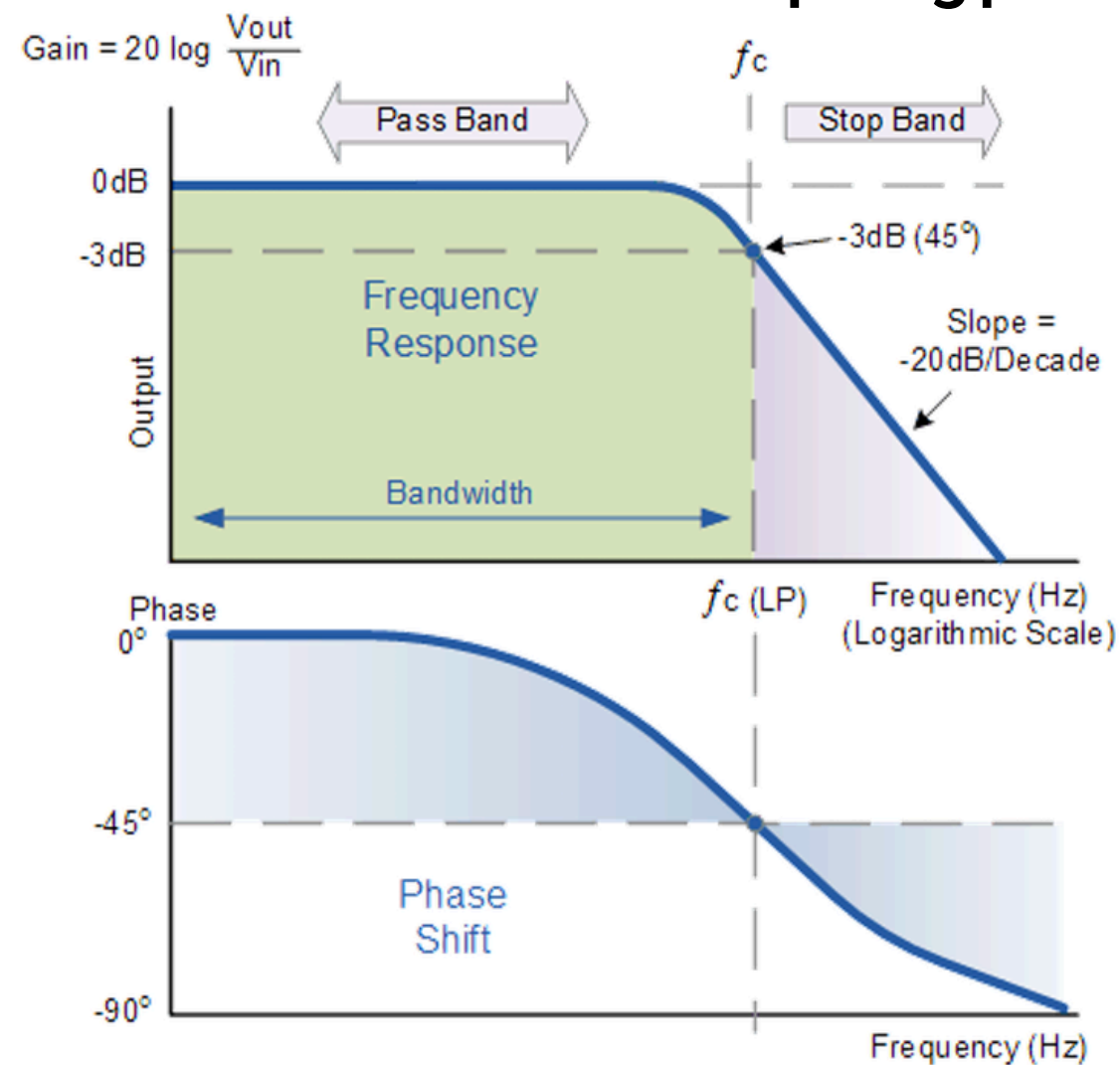
Why this behavior?

This is because the reactance of the capacitor is high at low frequencies and blocks any current flow through the capacitor.

Theory behind

Mathematical Formulas

Cut-off Frequency point



Mathematical model

$$f_c = \frac{1}{2\pi RC}$$

$$\text{phase shift } \varphi = -\arctan(2\pi RC)$$

Gain in Decibel

$$\text{Gain in dB} = 20 \log \frac{V_{out}}{V_{in}}$$

Theory behind

In our case: theoretical
Cutoff Frequency

Cutoff Frequency

$$f_c = \frac{1}{2 \pi R C}$$

R

$1k\Omega \pm 5\%$

C

$2.2 nF \pm 10\%$

$$f_c = \mathbf{106.4 kHz}$$

Theory behind

Uncertainty Model → Uncertainty propagation formula

$$u_c^2(y) = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \right)^2 u_i^2(x_i)$$

Why use the Uncertainty propagation formula ?

Uncertainty of cut off frequency is affected by the uncertainty of the involved elements (R and C).

Theory behind

Uncertainty Model → Uncertainty propagation formula

Variable	Symbol	Value
Resistance	R	$680\ \Omega$
Uncertainty in R	u_1	$20\ \Omega$
Capacitance	C	$2.20\ nF$
Uncertainty in C	u_2	$0.13\ nF$

Data

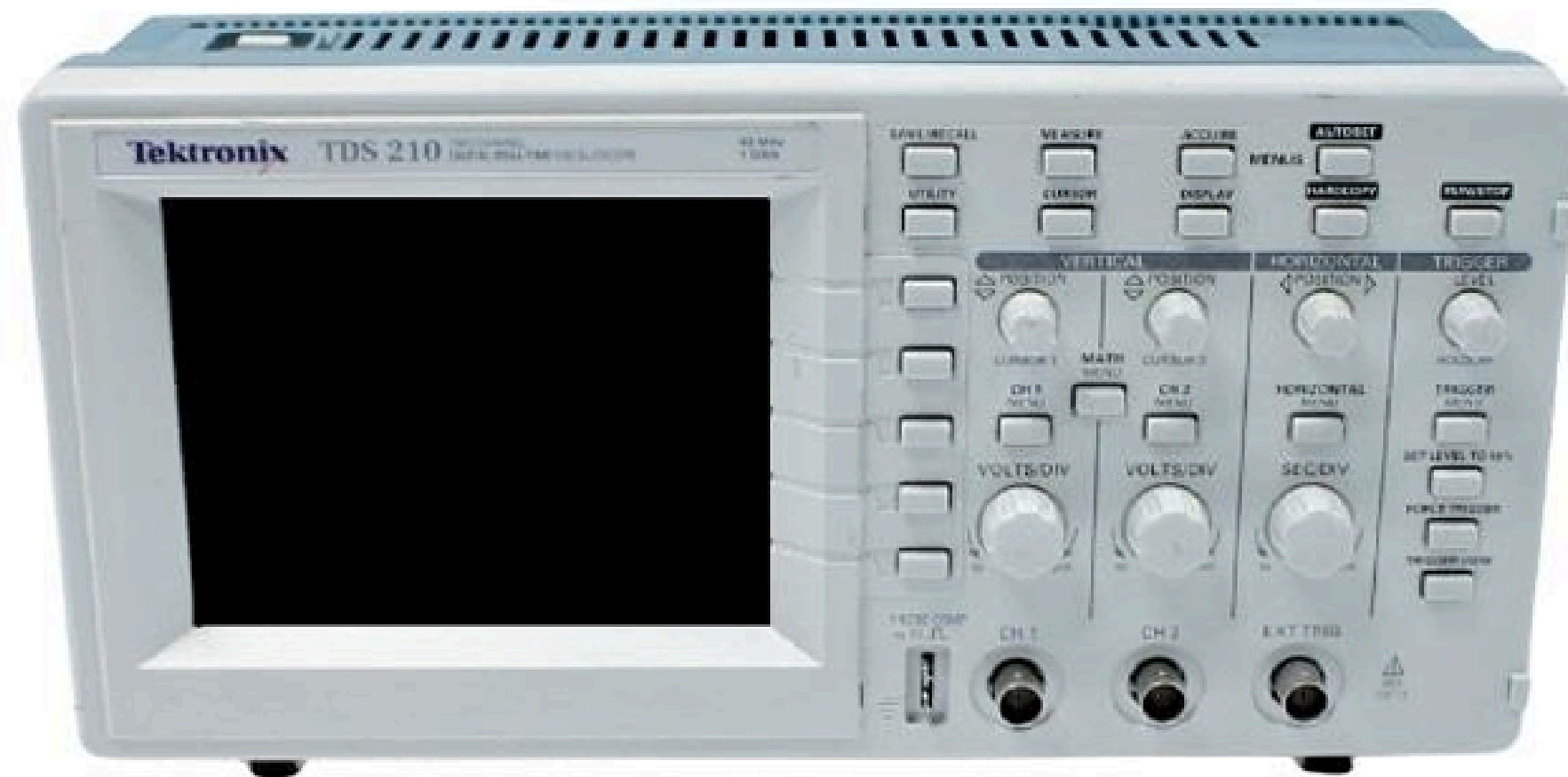
Theoretical Uncertainty

Evaluation process

Cutoff frequency: $\approx 106.4\ kHz$
Uncertainty: $uc(fc) \approx 6.9\ kHz$

What are the instruments?

Oscilloscope: Tektronix TDS 210



Bandwidth:

- 60 MHz (TDS 210)
- Selectable 20 MHz bandwidth limit
- Sample Rate: 1 GS/s
- Record Length: 2,500 points for each channel



What are the instruments?

Function Generator: Keysight 33220A



Key Features:

- 20 MHz Sine and Square waveforms
- Pulse, Ramp, Triangle, Noise, and DC waveforms
- 10 mVpp to 10 Vpp amplitude range
- AM, FM, PM, FSK, and PWM modulation types
- 14-bit, 50 MSa/s, 64 k-point arbitrary waveforms



How to link them?

One BNC-BNC coaxial cable



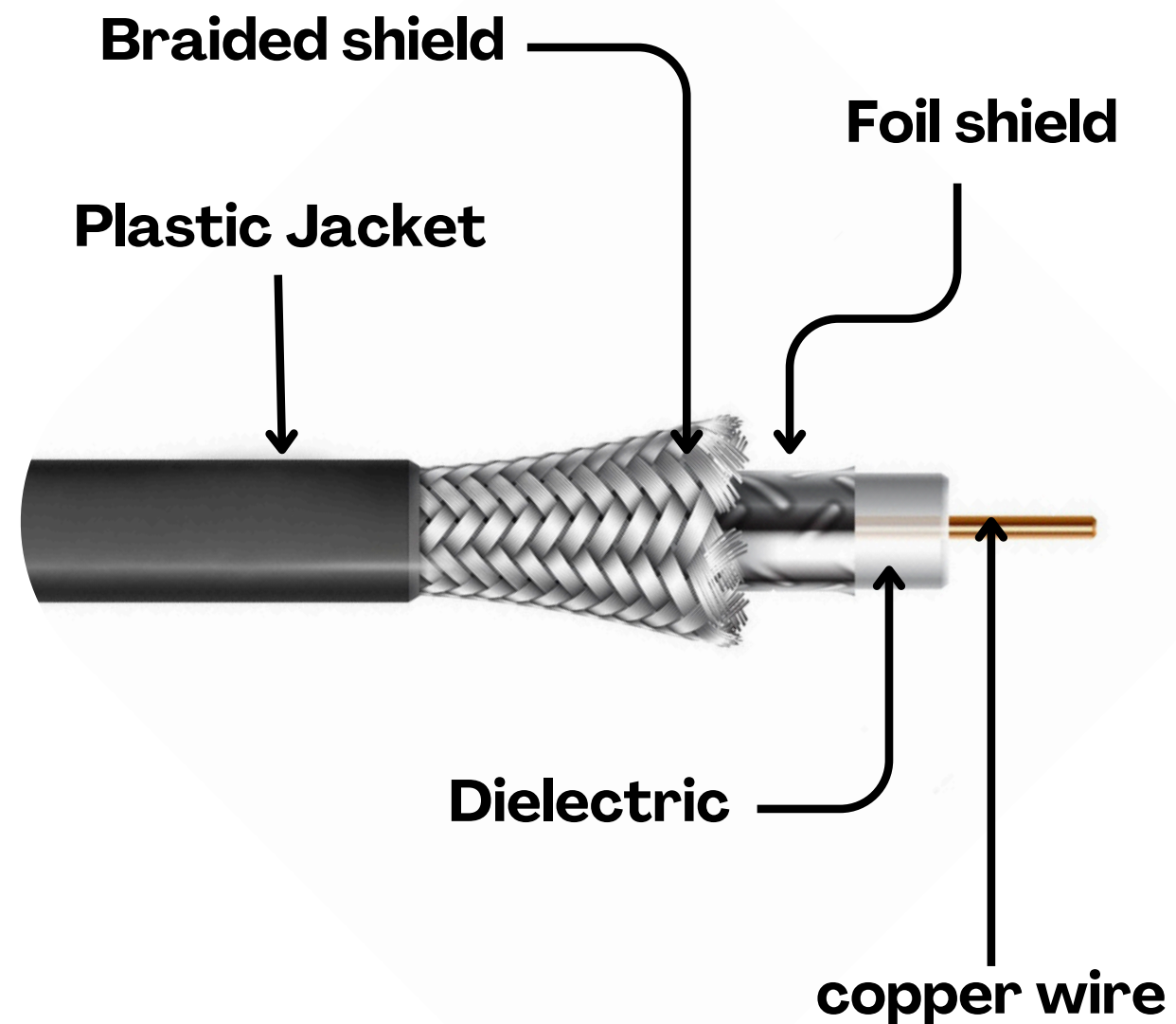
Why in parallel?

- Instruments share the same Voltage

Why a Coaxial Cable?

- Impedance closely matches the output impedance of the connected instruments
- Shield from external noise

How to link them?



Why the Plastic jacket?

- Shield from external interferences

Why the Braided shield?

- Prevent that the cable act as an antenna
- Protecting the signal from external noise
- Acting as a reference ground for the instruments.

How to link them?

Two BNC-Crocodile coaxial cable and T connector

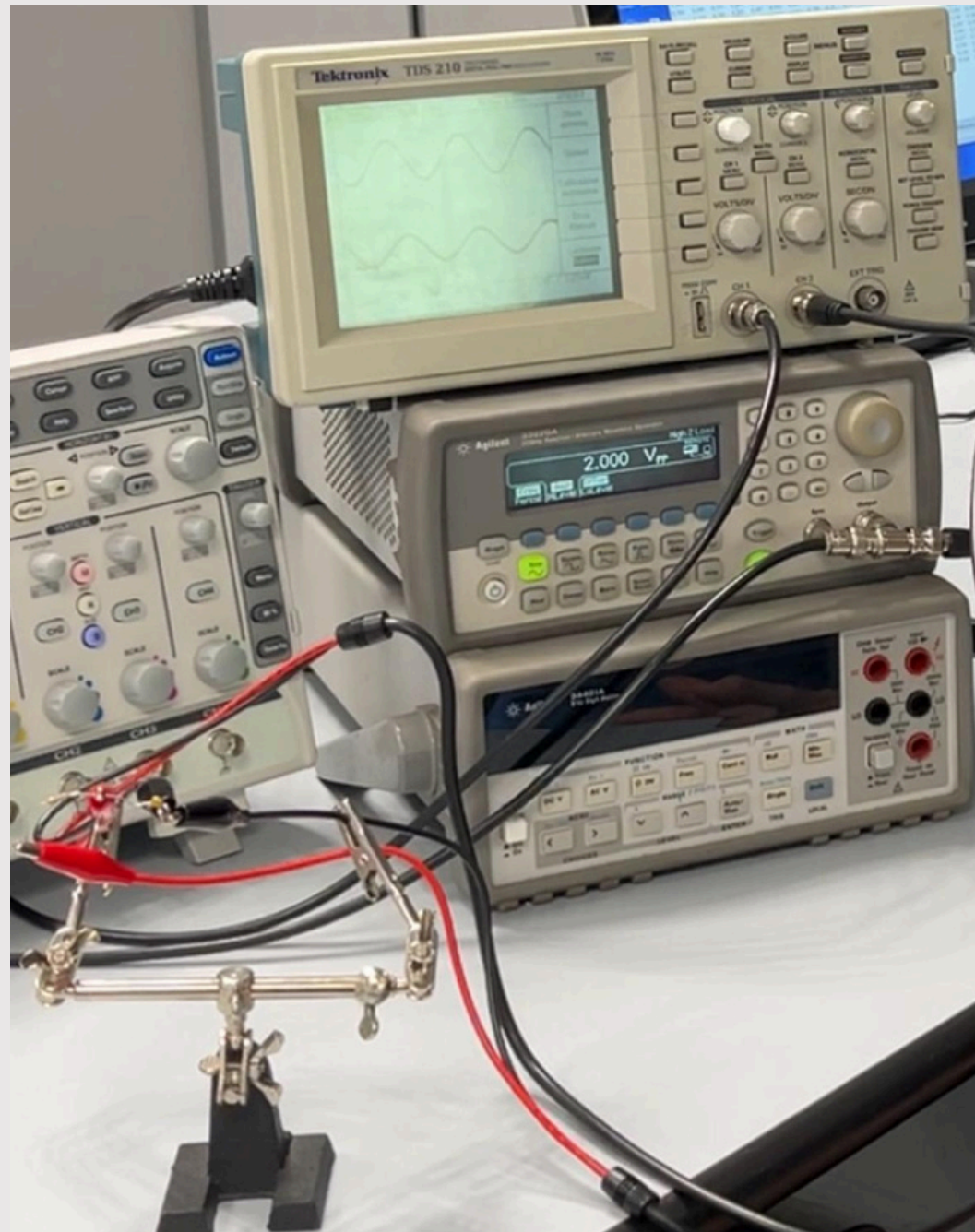


T connector



BNC-Crocodile
coaxial

How to link them?



Connect one coaxial cable between the T connector and DSO-CH1 of the oscilloscope

Connect one coaxial cable (BNC-Crocodile) between the DSO-CH2 and the RC filter output

Connect one coaxial cable (BNC-Crocodile) between the T connector and the RC filter input

Procedure

Set of the Function Generator

Sinusoidal Signal

f

[0 – 300] kHz
steps of 10kHz

0 V

Offset

A

2 V_{pp}

High
Impedance

Out-Imp

Procedure

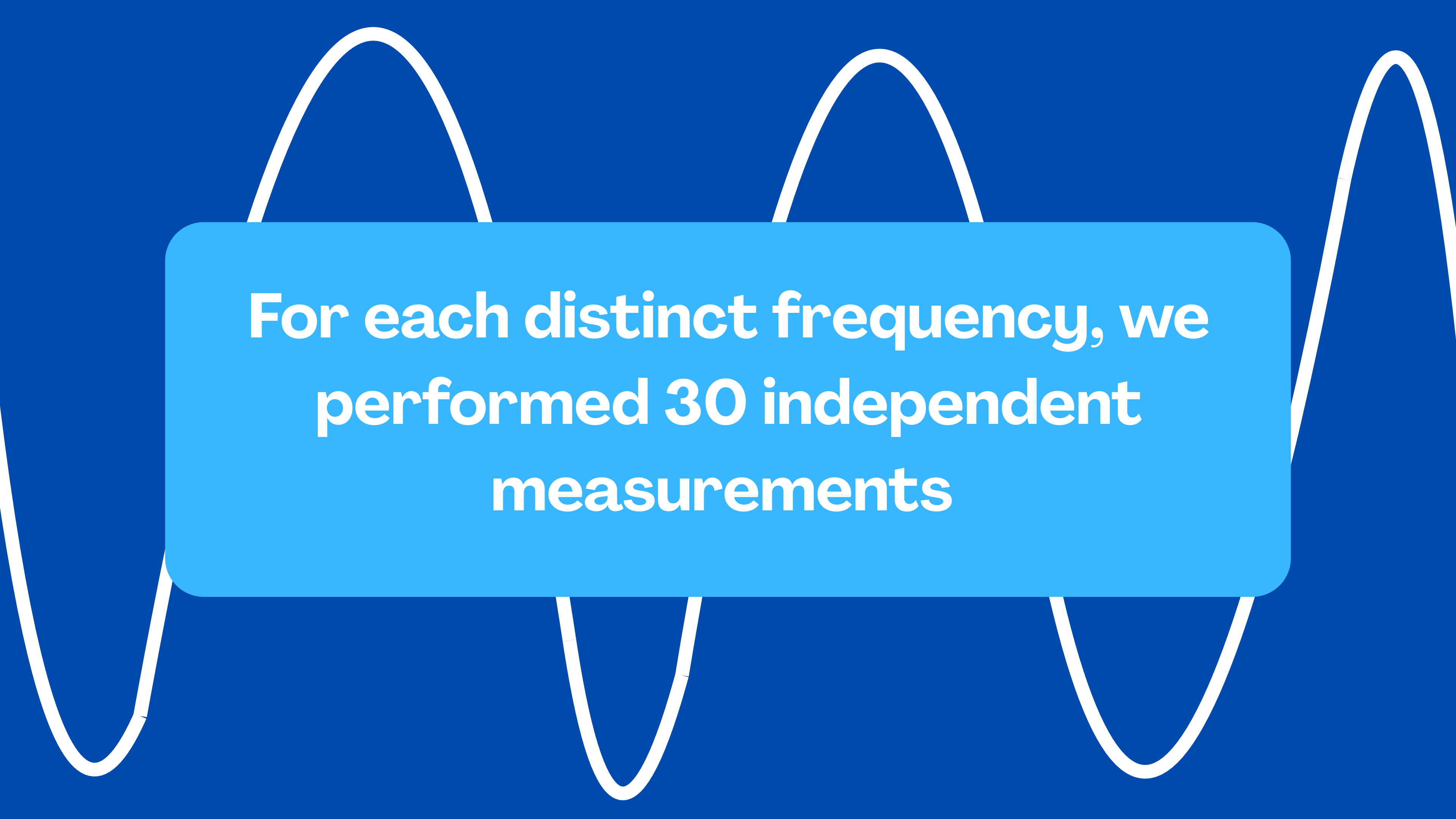
Evaluate the gain ratio

Gain ratio:

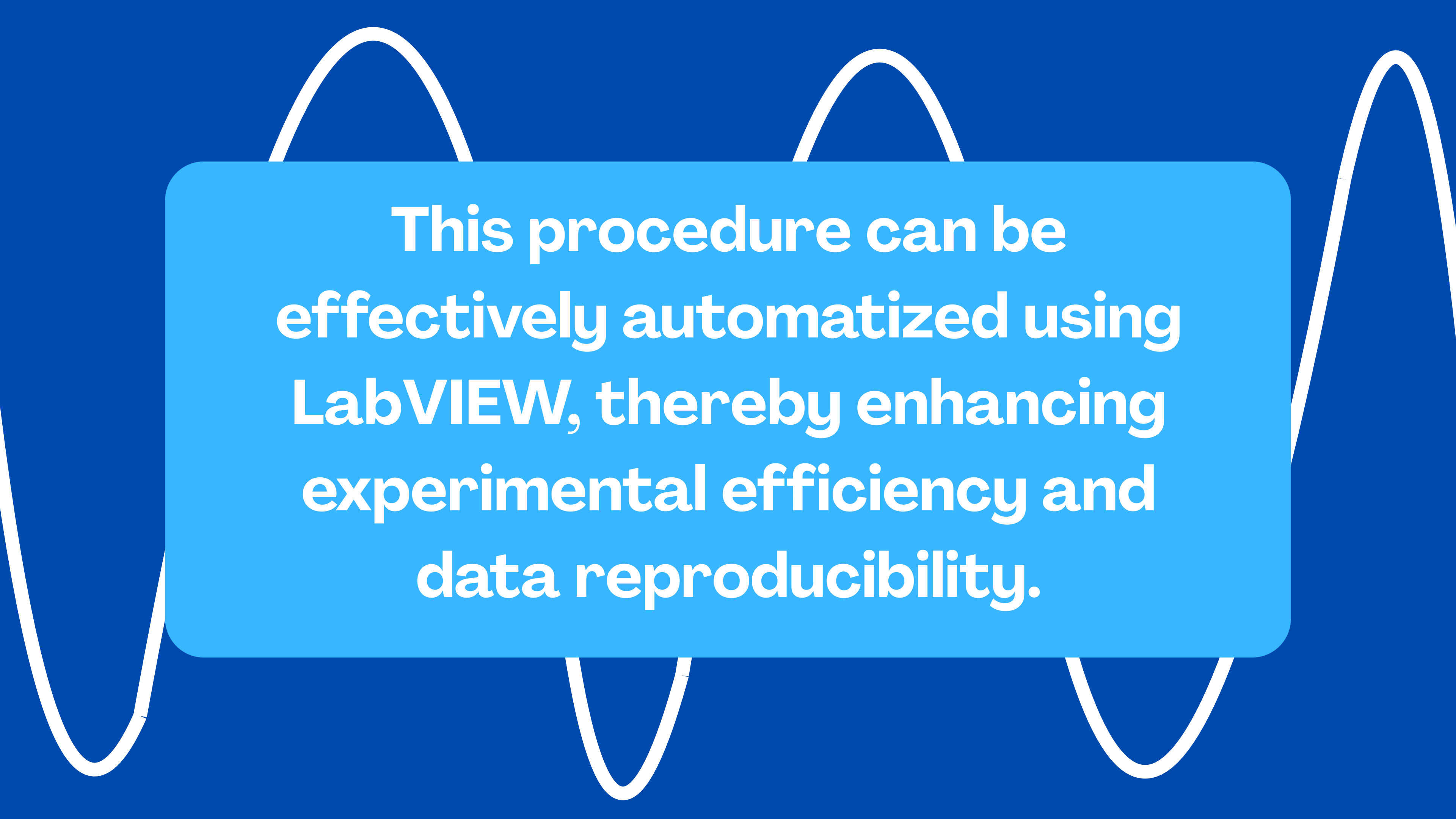
Measure the amplitude of the sinusoid on CH1, A1 and CH2, A2

The gain is the ratio $A2/A1$

It allows us to determine where is the cut-off frequency

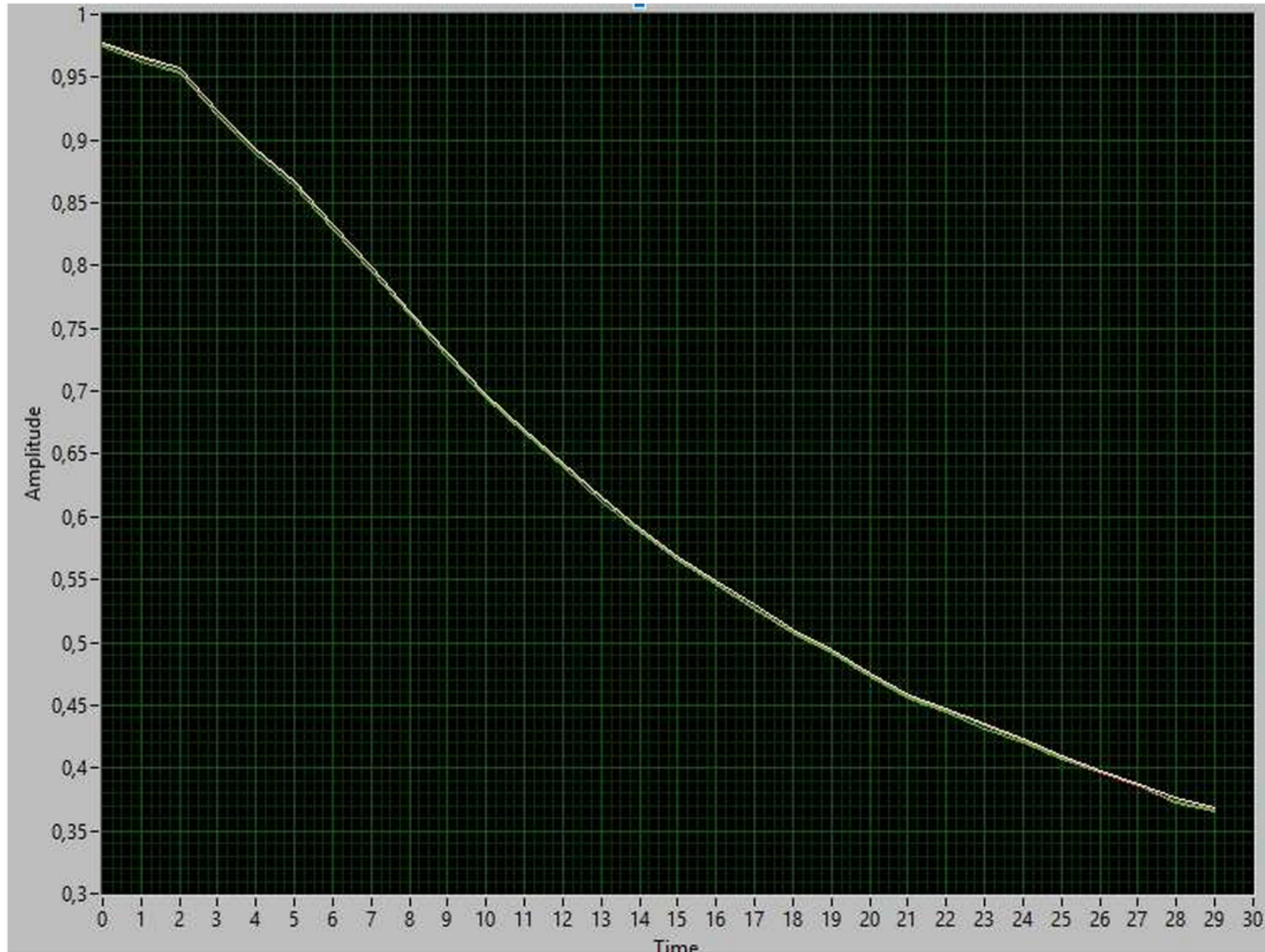


**For each distinct frequency, we
performed 30 independent
measurements**



**This procedure can be
effectively automatized using
LabVIEW, thereby enhancing
experimental efficiency and
data reproducibility.**

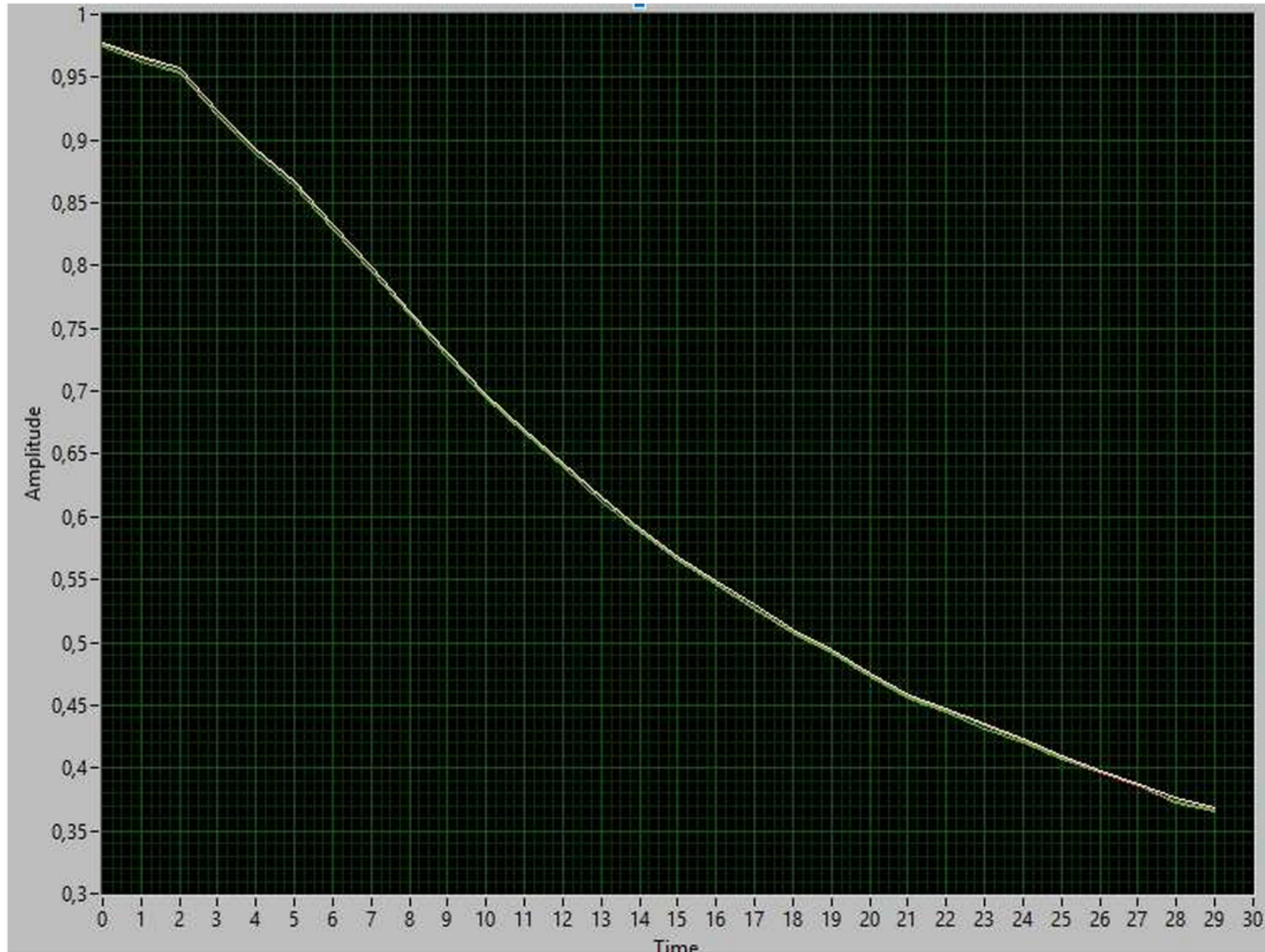
Experimental Evaluation



For each frequency
evalute :

- Mean gain value (30 values)
- Standard deviation (σ)
- Standard error of the mean

Experimental Evaluation



Low frequencies (1-50kHz)

- Mean gain $\approx 0.96-0.98$
- Standard deviation $\approx 0.01-0.02$
- Very consistent measurements

Near cutoff (160-200kHz)

- Mean gain $\approx 0.70-0.66$
- Standard deviation ≈ 0.02
- Slightly higher variability during transition

High frequencies (200-300kHz)

- Mean gain $\approx 0.36-0.40$
- Standard deviation ≈ 0.01

Experimental Evaluation

Cutoff Frequency Determination

Occurs where gain
drops to $1/\sqrt{2}$ of the
maximum gain (0.98)

Reference level:
 $0.98/\sqrt{2} \approx 0.69$

Conclusion

Experimental Vs Theoretical results

The theoretically calculated cutoff frequency (106.4 kHz) does not fall within the experimentally observed range (160-200 kHz). Do to:

- **Oscilloscope Input Impedance:** is high but not infinite
- **Cable Capacitance:** An increased total capacitance would lower the actual cutoff frequency

Conclusion

Hard Skills Acquired

- **Instruments Set up**
- **Evaluating the Cutoff Frequency of an RC LPF**
- **Uncertainty and cut-off frequency evaluation**

Conclusion

Hard Skills Acquired

- **Data Analysis and Interpretation**
- **Oscilloscope and Function Generator usage**

Conclusion

Soft Skills Developed

- **Experiment Organization**
- **Teamwork and Collaboration**
- **Communication**

Conclusion

Soft Skills Developed

- **Problem-Solving**
- **Critical Thinking**
- **Time Management**
- **Attention to details**

Thank you for your attention

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